14 Show that $\frac{p}{p-7} \div \frac{p^2}{p^2-49}$ can be written in the form $a + \frac{b}{p}$ where a and b are integers.

$$\frac{p}{p-7} \div \frac{p^2}{p^2-49} = \frac{p}{p-7} \times \frac{p^2-49}{p^2}$$

1 mark
Invert the SECOND fraction and multiply.

$$p^2 - 49 = (p-7)(p+7)$$

1 mark
Difference of squares.

$$\frac{p(p-7)(p+7)}{(p-7)p^2} = \frac{p+7}{p}$$

1 mark
Cancel common non-zero factors.

$$= 1 + \frac{7}{p}$$

1 mark
Split the numerator: a = 1 and b = 7.

Original expression requires $p \neq -7, 0, 7$.

(Total for Question 14 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA